

Temporal Degree Centrality Measures and Epidemic Dynamics in SIR-driven Adaptive Networks

Raffaele D'Agostino and Davide Volpi

Universitat Politècnica de Catalunya, Barcelona, Spain

<https://www.upc.edu/ca>

Abstract. Traditional static network analysis fails to capture the co-evolution of topology and dynamics in adaptive systems. This study investigates the interplay between temporal centrality measures and SIR epidemic spreading on networks where susceptible nodes actively sever links with infected neighbours. We introduce a memory-based Temporal Momentum Centrality to quantify the inertia of connectivity changes. Through Monte Carlo simulations, we demonstrate that adaptive rewiring creates a "dynamic quarantine" effect, significantly suppressing the epidemic threshold compared to static benchmarks. We show that standard Heterogeneous Mean-Field (HMF) theory overestimates the effective reproduction number, $R(t)$, as it neglects the local correlations generated by targeted isolation. Furthermore, we identify an "inversion of centrality": while high static degree is necessary for seeding, the temporal loss of connectivity serves as the strongest dynamic signature of a successful spreader. These findings suggest that monitoring the rate of degree collapse is essential for identifying active transmission in adaptive networks.

Keywords: Adaptive Temporal Networks · Temporal Centrality Measures · SIR Epidemic Model

1 Introduction

1.1 Context and Motivation

Network science has traditionally relied on static graph representations to model complex systems, effectively aggregating interactions over time into a single set of edges. While this approach has yielded significant insights into the structural properties of systems ranging from social interactions to biological pathways, it discards the temporal dimension of connectivity. Real-world networks, however, are dynamic: interactions occur at specific times, have durations, and often exhibit bursty or periodic patterns.

The shift from static to *temporal network* analysis is crucial because the timing and ordering of events can fundamentally alter the outcome of dynamic processes. Traditional analysis often relies on static snapshots—instantaneous views of the network frozen at a specific time step. However, this approach fails

to capture the continuous flow of causality between time steps. A dynamic process, such as an epidemic, does not propagate solely within a single snapshot; it survives and spreads across the temporal sequence of these snapshots. Consequently, analysing isolated instantaneous states may miss the critical "bridging" events that allow a contagion to persist and expand over time, leading to inaccurate predictions of diffusion speed and outbreak size.

This project specifically investigates these discrepancies in the context of *adaptive temporal networks*, where the topology is not merely a fixed background for dynamics but an evolving structure that rewires in response to the spreading process itself. By analysing two distinct dynamic processes, temporal degree centrality with memory and epidemic spreading on rewired networks, we aim to capture the relation between network evolution and contagion dynamics.

1.2 Problem Statement

The core objective of this research is to quantify the differences between static and temporal network analysis. We focus on how the time evolution affects node importance and disease propagation dynamics.

Module A: Dynamic Metric Analysis Centrality measures identify the most important nodes in a network. We focus on *Degree Centrality*, which measures a node’s immediate influence through its number of connections. While static Degree Centrality simply counts the total number of unique partners over the entire dataset, it fails to capture the intermittency of interactions. In a temporal context, a node may appear to be a "hub" in the aggregate view despite being inactive for long periods. This module examines *Temporal Degree Centrality*, specifically investigating *Temporal Degree Centrality* metrics incorporating a time-decaying memory window to address the following questions:

- Impact of Temporal Memory: how does the introduction of a time-decaying memory window alter centrality rankings compared to static degree centrality?
- Sensitivity to Time Scale: how does varying the length of this memory window amplify the differences between nodes with stable, continuous interactions versus those with sporadic, bursty activity?

Module B: Epidemic Modeling The temporal structure of contacts is particularly critical for spreading processes. We analyze an epidemic model (SIR) to understand how the time-varying nature of edges impacts the epidemic threshold (R_0) and final outbreak size. A key focus is testing the hypothesis by Pastor-Satorras et al., which suggests that the effective reproduction number $R(t)$ can be estimated using the instantaneous degree distribution of the network at time t . The temporal ordering of contacts fundamentally constrains spreading paths. We simulate an SIR epidemic process on a synthetic rewired temporal network to answer:

- Thresholds and Outbreak Size: Is the epidemic threshold (R_0) higher, and the final outbreak size smaller, in temporal networks compared to static equivalents, due to the disruption of contact paths over time?
- Estimation of $R(t)$: Can the effective reproduction number $R(t)$ be accurately approximated using the instantaneous degree distribution at each time step, as suggested by Pastor-Satorras et al. [1]?
- Influence of Initial Conditions: How do the size of the outbreak differ when the infection is initiated by nodes with high versus low temporal centrality? Is there a correlation between the outbreak size and the analysed Temporal Degree Centralities?

Report structure This report is organized as follows to address the research questions outlined in the proposal. First, we present the quantitative *Results* obtained from the simulations and the relative *Discussion* providing a critical interpretation of the findings, linking the observed phenomena to theoretical frameworks. Finally, the *Methods* section details the methodological framework: computational models, algorithms, justifications, statistical framework used to obtain the results. Detailed implementation and parameter settings used to generate the data, ensuring the reproducibility of the study, are given in Appendix A.

2 Results and Discussion: Temporal Dynamics in Adaptive Networks

In this section, we present the analysis of the proposed *Temporal Momentum Centrality*, denoted as $M(t)$, in comparison with the static instantaneous degree $k(t)$ and the simple window average $\bar{k}(t)$. Rather than claiming statistical representativeness from a small sample, we use a limited set of randomly selected nodes only as an illustrative visualization to make the underlying mechanism readable at the level of single-node trajectories. The qualitative patterns discussed below (growth/decay with inertia and the dependence on the memory window W) are theoretically expected from the definition of a memory-based momentum metric (Section 5) and from the adaptive rewiring rule, and the plotted traces serve to demonstrate how these mechanisms manifest in concrete realizations of the process.

Figure 1 illustrates the temporal evolution of these metrics alongside key epidemic events: the moment of Infection (vertical dashed line) and Recovery (vertical dotted line).

2.1 Modeling Growth and Decay Phases (H1)

The time series panels (Columns 2–6) confirm that $M(t)$ captures structured phases in node importance during the epidemic, rather than only reacting to instantaneous edge fluctuations. Across the sampled nodes, this supports Hypothesis H1 by highlighting how temporal accumulation and momentum give

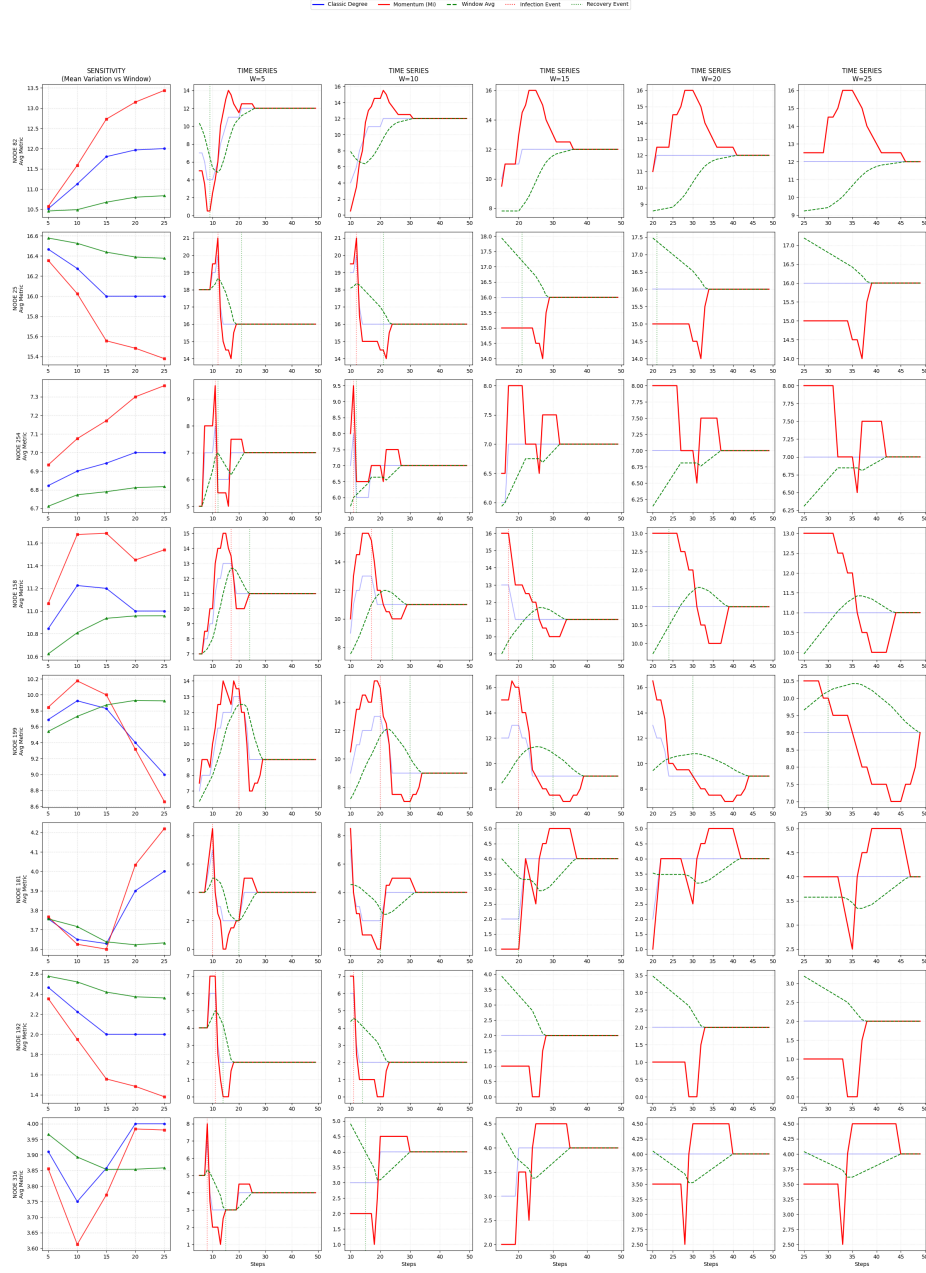


Fig. 1. Detailed tracking of 8 randomly sampled nodes within the adaptive SIR network with parameters $\beta/\gamma = 1,625$, $w = 0.3$, $p_{rand} = 0.2$. The first column presents the sensitivity analysis, reporting the variation of the mean metric value as a function of the memory window size W . Columns 2–6 depict the instantaneous evolution of the metrics for increasing window sizes ($W \in \{5, 10, 15, 20, 25\}$). The plots compare the proposed Momentum Centrality $M(t)$ (red solid line) against the static Instantaneous Degree $k(t)$ (light green line) and the simple Window Average $\bar{k}(t)$ (blue line). Vertical markers indicate epidemic status changes: *Infection* (dashed line) and *Recovery* (dotted line).

more importance to the growth/decay status of a node that is not evident from $k(t)$ alone.

- **Growth Phase (Pre-Infection).** For several nodes, the pre-infection period is characterized by a rapid accumulation of contacts. The temporal momentum metric $M(t)$ *amplifies* the growth trend of the static degree $k(t)$. Because the momentum component is sensitive to the positive rate of change in connectivity, $M(t)$ acts as a signal booster, consistently overshooting the instantaneous degree ($M(t) > k(t)$) during the rise. This behaviour makes $M(t)$ particularly effective at highlighting nodes that are not just highly connected, but are gaining importance with high *velocity*, distinguishing active acquisition of edges from static stability.
- **Decay Phase and Inertia (Post-Infection).** Following infection, the adaptive rewiring mechanism typically causes a sharp drop in the static degree $k(t)$ as neighbours disconnect. The momentum metric $M(t)$ reacts to this negative gradient by exaggerating the decline. Rather than decaying gradually, $M(t)$ exhibits a steep downward trajectory that may transiently undercut the static degree, reflecting the intensity of the connection loss. This allows the metric to distinguish between a simple low-degree state and a *rapid collapse* in centrality, effectively encoding the "force" of the node's isolation in the post-infection phase.

2.2 The Memory Window as a Scale Parameter (H2)

For small memory windows (e.g., $W = 5$), the metric $M(t)$ is highly sensitive to the instantaneous rate of change. As observed in the time series (Column 2), the trajectory exhibits sharp fluctuations, reacting aggressively to short-lived rewiring bursts or transient contact spikes. While this provides high temporal resolution, it risks amplifying high-frequency noise, making it difficult to distinguish between a structural trend (e.g., a sustained epidemic spread) and random topological volatility.

As W increases (e.g., $W \geq 20$), the metric acts less as a volatility tracker and more as a trend integrator. The visible effect is not merely smoothing, but the emergence of *temporal inertia*: peaks broaden into sustained plateaus and rapid drops are buffered by the memory of past connectivity. This allows $M(t)$ to filter out local noise and highlight dominant *structural phases*, such as the prolonged accumulation of risk prior to infection, that are otherwise obscured by the jaggedness of the static degree.

The sensitivity analysis (Column 1) reveals that increasing W fundamentally shifts the metric's baseline depending on the node's history. For nodes with a dominant growth phase (e.g., Node 82), a larger W accumulates more positive momentum, driving the average metric value *upward*. Conversely, for nodes in decline (e.g., Node 25), a longer memory emphasizes the loss of connections, pulling the average *downward*. Operationally, this confirms that choosing W is a trade-off: small windows detect "micro-events" (instantaneous changes), while large windows characterize the "momentum" of the epidemic progression.

Finally, a critical limitation emerges when the memory window W exceeds the duration of the transient event itself. This is clearly visible in Node 192 at $W = 25$: the node undergoes a rapid cycle of growth followed immediately by a decay (infection and isolation). Because both the rise (positive momentum) and the fall (negative momentum) fall within the same integration window, they effectively cancel each other out. Instead of highlighting the volatility, the metric aggregates these opposing dynamics into a flattened trajectory, obscuring the epidemic "pulse." This suggests that while a larger W filters noise, an excessively long window relative to the process timescale destroys temporal resolution, rendering the metric blind to short but intense structural changes.

3 Results and Discussion: SIR Epidemic Model in Adaptive Networks

3.1 Epidemic Dynamics with Dual Rewiring Mechanisms (H3, H4)

To verify hypotheses H3 and H4, we simulated the SIR epidemic process on a temporal adaptive network and compared it against a static benchmark with equivalent connectivity. Figure 2 reports the Final Epidemic Size ($N_{infected}/N_{total}$) as a function of the Transmission Rate Ratio (β/γ).

The simulation results provide strong support for hypotheses H3 and H4, demonstrating that the adaptive nature of the network acts as an effective barrier to disease propagation. The interplay between the spreading rate (β), the protective adaptive rewiring ($w = 0.3$), and the background stochastic noise (p_{rand}) generates distinct dynamical regimes compared to the static baseline.

The static network (black dashed line) transitions to a global epidemic state almost immediately ($\beta/\gamma \approx 0$), a behaviour typical of heterogeneous static networks where hubs facilitate rapid percolation. In contrast, the temporal network exhibits a marked resistance to spreading, shifting the critical epidemic threshold to $\beta/\gamma \approx 0.4$. This delay confirms H3 and indicates that the adaptive mechanism functions as a "dynamic quarantine" [3]: susceptible nodes actively break connections with infected neighbours faster than the pathogen can traverse them. For an outbreak to occur, the infection rate must be sufficiently high to outpace this severance of transmission pathways, effectively raising the minimal R_0 required for percolation. Even beyond the critical threshold, the final epidemic size in the temporal network is consistently lower than in the static case, validating H4. While the static topology offers permanent paths for the virus to reach the entire giant component, the adaptive rewiring fragments the temporal giant component. The combined effect of cutting infectious links (adaptive strategy) and reshuffling others (p_{rand}) isolates "pockets" of susceptible individuals. These nodes become topologically unreachable before the infection can access them, capping the total saturation of the population. A key feature of the results is the large standard deviation (red shaded area) observed in the transition region ($\beta/\gamma \in [0.5, 1.0]$). In this critical regime, the outcome is binary and highly unstable. Since the infection rate is barely higher than the rewiring rate, the fate

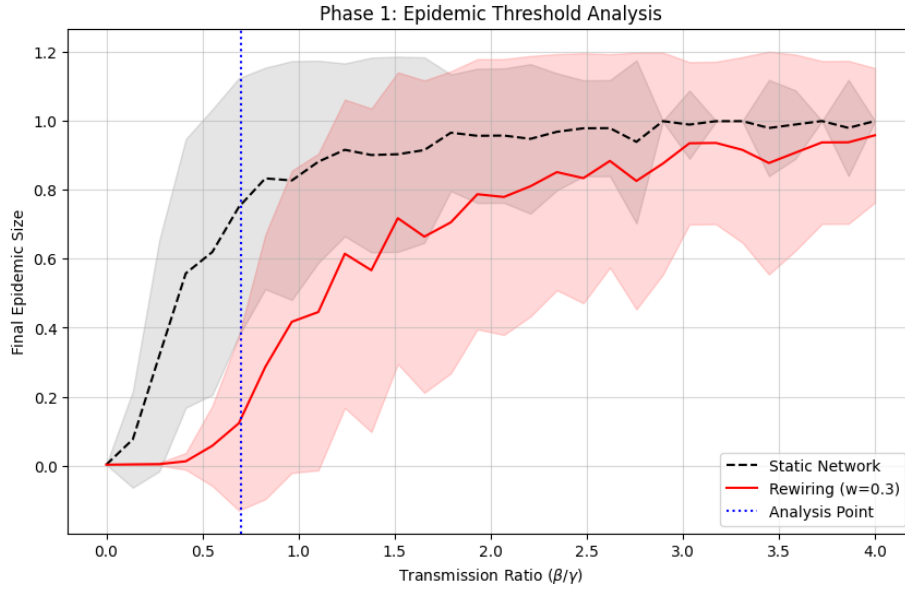


Fig. 2. Final epidemic size as a function of the transmission ratio β/γ for the static network (black dashed line) and the adaptive temporal network with rewiring probability $w = 0.3$ (solid red line). Shaded regions represent the standard deviation across simulations. The vertical blue dotted line indicates the specific transmission ratio selected for the subsequent analysis of initial node centrality (H6).

of the epidemic depends on the very first interactions: if the initial neighbours manage to disconnect just in time, the outbreak dies out immediately; if the virus succeeds in infecting just one or two neighbours before the links are cut, it can escape this initial containment and spread widely. This bifurcation leads to high variance across simulation runs.

3.2 Temporal Evolution of the Effective Reproduction Number $R(t)$

To address hypothesis H5, we examined the validity of estimating the effective reproduction number using instantaneous topological properties. Figure 3 presents the comparative analysis between the theoretical prediction of [1] and the realized spreading potential in the adaptive network ($w = 0.3$).

The analysis tracks two diverging metrics:

- Theoretical Estimation (HMF): The dashed blue line depicts the Heterogeneous Mean-Field approximation, $R_{HMF}(t)$, derived from the time-varying moments of the degree distribution:

$$R_{HMF}(t) = \frac{\beta}{\gamma} \left(\frac{\langle k^2(t) \rangle}{\langle k(t) \rangle} - 1 \right) \quad (1)$$

This metric assumes that the network is “annealed” or well-mixed at every time step, implying that the instantaneous degree distribution alone dictates the spreading potential [1].

- Effective Reproduction Number (Microscopic): The solid red line represents the actual average number of secondary infections caused by infected nodes at each time step during the simulation.

The results indicate a fundamental breakdown of the theoretical prediction in the adaptive regime. While both metrics show a decaying trend characteristic of epidemic depletion, the effective $R(t)$ is consistently and significantly lower than the Theoretical HMF prediction. The divergence is most pronounced in the early phase ($t < 10$), where the theory predicts a massive spreading potential ($R \approx 23$), whereas the actual adaptive process limits this to $R \approx 13$. Furthermore, the effective $R(t)$ breaches the critical threshold $R = 1$ faster than the theoretical curve, demonstrating again that the adaptive rewiring suppresses the epidemic more efficiently than topological statistics alone would suggest.

The discrepancies observed in Figure 3 provide strong evidence refuting the direct applicability of hypothesis H5 for adaptive networks. The failure of the theoretical model can be attributed to the invalidity of the “annealed network” assumption [1] in the presence of targeted rewiring.

The Heterogeneous Mean-Field (HMF) framework assumes that connections are reshuffled randomly and rapidly, effectively destroying local correlations. However, the adaptive rewiring mechanism (w) is not a random shuffling process but a targeted reaction that generates persistent local correlations:

1. Dynamic Local Fragmentation: as susceptible nodes sever links with infected neighbours, they create a local “quarantine bubble” around the infected

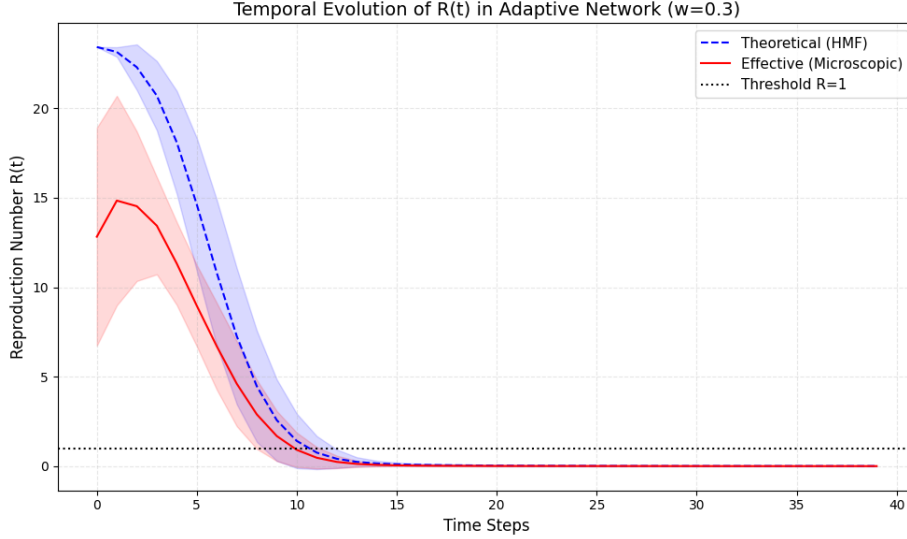


Fig. 3. Comparison of the theoretical reproduction number derived from the instantaneous degree distribution (Blue Dashed) and the effective reproduction number measured from microscopic simulation (Red Solid). Shaded areas indicate standard deviations.

nodes. This leads to a local depletion of susceptibles that is far more severe than the global average suggests. The infected nodes retain high degrees in the global distribution, but their specific neighbours are increasingly immune or recovered, neutralizing their actual spreading capability.

2. Inefficiency of Random Mixing: The background random rewiring (p_{rand}) operates on a timescale too slow to counteract this targeted isolation. The system effectively freezes into local protective configurations that the mean-field “annealed” assumption, which averages over all possible configurations, fails to capture.
3. Violation of Mass Action: HMF theory relies on the mass action law, assuming the force of infection is proportional to the global density of infected links. In our adaptive model, the specific $S-I$ links required for transmission are selectively destroyed. Consequently, the global moments $\langle k \rangle$ and $\langle k^2 \rangle$ become misleading proxies for transmission risk, as they no longer reflect the connectivity available to the pathogen.

In conclusion, while the instantaneous degree distribution captures the network’s potential connectivity, it fails to account for the functional isolation of infected nodes. The effective $R(t)$ is suppressed because the adaptive mechanism dynamically depletes the specific pathways the virus requires, a microscopic correlation that persists despite global random fluctuations.

3.3 Results of the Meta-Analysis (H6)

To test hypothesis H6, we analysed the statistical association between the temporal evolution of node centrality and the final epidemic size across 32 validated seed nodes. The analysis was repeated for three observation windows ($W \in \{3, 6, 9\}$) to assess robustness against the timescale of the epidemic peak.

Descriptive Analysis of Correlations Table 1 summarizes the distribution of Spearman correlation coefficients (ρ) obtained from the node-level simulations. The results reveal a consistent negative association across all metrics. *Average Degree* exhibits the strongest correlation ($\bar{\rho} \approx -0.65$), indicating that nodes maintaining high connectivity during the window are strongly associated with smaller outbreaks. *Momentum* also shows a negative mean correlation, though with higher variance ($\sigma \approx 0.31$), suggesting a more heterogeneous response across different node types.

Table 1. Descriptive Statistics of Correlation Coefficients (ρ). The table aggregates the Spearman correlations calculated for each seed node. A negative Mean/Median ρ indicates that higher values of the metric are associated with smaller epidemic sizes.

Metric	Window $W = 3$			Window $W = 6$			Window $W = 9$		
	Mean ρ	Median ρ	Std σ	Mean ρ	Median ρ	Std σ	Mean ρ	Median ρ	Std σ
Momentum	-0.326	-0.204	0.307	-0.287	-0.168	0.312	-0.268	-0.149	0.306
Avg Degree	-0.645	-0.669	0.193	-0.656	-0.683	0.190	-0.666	-0.697	0.185
Deg at Peak	-0.370	-0.350	0.172	-0.370	-0.350	0.172	-0.370	-0.350	0.172

Statistical Significance and Normality Checks To validate these observations, we performed a rigorous statistical assessment of the correlation distributions. As shown in Table 2, the Shapiro-Wilk test indicates that the correlation coefficients for *Momentum* are not normally distributed ($p < 0.05$ across all windows), necessitating the use of non-parametric tests.

The significance of the negative correlation is confirmed by all three statistical tests (T-Test, Wilcoxon Signed-Rank, and Permutation Test), which yield p-values well below the significance threshold ($\alpha = 0.05$). The convergence of these tests provides strong evidence that the observed negative correlations are systematic and not artifacts of the data distribution.

3.4 Discussion: Topological Feedback as a Signature of Spreading

The results from the 32 validated seed nodes indicate a strong negative correlation between centrality maintenance and epidemic size. To interpret this correctly, one must account for the filtering process applied to the 18 excluded nodes were primarily low-degree seeds where the epidemic failed to propagate

Table 2. Statistical Significance Tests. The Shapiro-Wilk test checks the normality of the ρ distribution. Since Momentum violates normality ($p < 0.05$), the non-parametric Wilcoxon and Permutation tests provide the most reliable confirmation of significance. All p-values $< 10^{-4}$ confirm the correlations are significantly different from zero.

Metric	Window $W = 3$				Window $W = 6$			
	Shapiro p	T-Test p	Wilcox p	Perm p	Shapiro p	T-Test p	Wilcox p	Perm p
Momentum	0.007	$< 10^{-4}$	$< 10^{-4}$	$2 \cdot 10^{-4}$	0.017	$< 10^{-4}$	$< 10^{-4}$	$2 \cdot 10^{-4}$
Avg Degree	0.133	$< 10^{-4}$	$< 10^{-4}$	$2 \cdot 10^{-4}$	0.106	$< 10^{-4}$	$< 10^{-4}$	$2 \cdot 10^{-4}$
Deg at Peak	0.656	$< 10^{-4}$	$< 10^{-4}$	$2 \cdot 10^{-4}$	0.656	$< 10^{-4}$	$< 10^{-4}$	$2 \cdot 10^{-4}$

Metric	Window $W = 9$			
	Shapiro p	T-Test p	Wilcox p	Perm p
Momentum	0.010	$< 10^{-4}$	$< 10^{-4}$	$2 \cdot 10^{-4}$
Avg Degree	0.096	$< 10^{-4}$	$< 10^{-4}$	$2 \cdot 10^{-4}$
Deg at Peak	0.656	$< 10^{-4}$	$< 10^{-4}$	$2 \cdot 10^{-4}$

beyond the first few steps, resulting in zero variance in both topological and epidemic metrics.

Consequently, the observed negative correlation in the remaining 32 active nodes captures the dynamics of *successful* outbreaks. In the adaptive regime ($w = 0.3$), the spread of infection triggers an immediate local response:

1. **Degree Loss as a Reaction:** When a seed node successfully transmits the pathogen to its neighbours, those neighbours, upon becoming infected, trigger the rewiring mechanism (rate w) to isolate themselves or are disconnected by their own susceptible neighbours.
2. **The Inverse Link:** Therefore, a sharp decrease in degree (lowering momentum) and a lower average degree over the window are not causes of the epidemic, but *consequences* of a successful expansion. A node that maintains its initial high degree (momentum $\approx$ static degree) indicates a failure to trigger this chain reaction, essentially signalling that the pathogen remained contained within the immediate neighbourhood without forcing a massive topological restructuring.

In conclusion, while high static centrality is a prerequisite for potential spreading (as shown by the need to filter out low-degree nodes), the *temporal* loss of that centrality serves as a dynamic signature that the potential has been realized into an actual outbreak.

4 Final Discussion and Conclusions

This study investigated the interplay between temporal network metrics and epidemic spreading on adaptive networks. By introducing a memory-based centrality metric, *Temporal Momentum Centrality* $M(t)$, and simulating an SIR-Adaptive process, we addressed the limitations of static topological analysis in

capturing the feedback loop between disease propagation and network restructuring.

4.1 Summary of Findings

Our results validate the proposed hypotheses across three distinct dimensions:

1. **Metric Dynamics (H1, H2):** The *Momentum Centrality* successfully captures the inertial properties of node connectivity. Unlike the instantaneous degree, which fluctuates rapidly, $M(t)$ highlights structural phases of growth and decay. The sensitivity analysis confirms that the memory window W acts as a scale parameter: short windows detect volatility, while longer windows integrate the trend, effectively filtering local noise but potentially obscuring rapid pulse-like events.
2. **Failure of Mean-Field Theory (H3, H4, H5):** We demonstrated that adaptive rewiring fundamentally alters the epidemic threshold. The network exhibits a "dynamic quarantine" effect, suppressing the final epidemic size significantly below the static benchmark. Crucially, we showed that standard Heterogeneous Mean-Field (HMF) theory overestimates the reproduction number $R(t)$ by nearly a factor of two in the early phase. This failure stems from the "annealed" assumption, which ignores the persistent local correlations (the "protective bubbles") created by the targeted rewiring of susceptible nodes.
3. **The Inversion of Centrality (H6):** Perhaps the most counter-intuitive finding is the relationship between centrality and spreading. While static hubs are necessary to ignite the process, their *temporal* trajectory reveals an inverse relationship: a rapid loss of centrality (negative momentum) is strongly correlated with larger outbreaks. In an adaptive network, a "crashing" degree is not a sign of weakness, but the dynamic signature of a successful spreader that has forced its neighbors to disconnect.

4.2 Conclusions

This work highlighted the critical limitations of applying static network theory to adaptive epidemic models. Our simulations quantified a substantial divergence between the theoretical Heterogeneous Mean-Field prediction and the realized spreading potential, confirming that the "annealed" approximation breaks down when topological changes are targeted rather than random.

Furthermore, we showed that the traditional interpretation of centrality is inverted in the adaptive regime. Our meta-analysis identified the *Windowed Average Degree* as the strongest statistical correlate of epidemic size ($\bar{\rho} \approx -0.65$), confirming that nodes which successfully propagate the infection are those that fail to maintain their connectivity over time. While less strongly correlated, the *Momentum Centrality* provides the mechanistic explanation for this phenomenon: it detects the specific *rate of collapse* ("negative momentum") that characterizes the adaptive isolation of a spreader.

These findings suggest that monitoring the temporal stability of node degree—specifically looking for sudden drops or sustained low averages in high-degree nodes—can serve as an effective proxy for identifying active spreading events in adaptive networks.

5 Methods

5.1 Temporal Network Representation

To model the evolution of a network that reasonably approximates social interactions, we choose as the initial graph G_0 a synthetic network generated by the Barabási–Albert (BA) model. The BA mechanism combines network growth with preferential attachment, meaning that newly added nodes are more likely to connect to already well-connected vertices, naturally producing the presence of highly connected hubs. In the asymptotic regime, this process yields a heavy-tailed (often referred to as scale-free) degree distribution, approximately following $P(k) \sim k^{-3}$, making G_0 a suitable starting point for studying temporal centrality metrics and dynamical spreading processes on structurally heterogeneous networks.

To model the evolving interactions within the system, we represent the temporal network as a discrete sequence of static graph snapshots. We denote the temporal network as $\mathcal{G} = \{G_0, G_1, \dots, G_T\}$, where each snapshot $G_t = (V, E_t)$ represents the state of the network at time step t . The node set V remains constant throughout the simulation (with size $N = |V|$), while the edge set E_t evolves dynamically according to rewiring rules defined together with our epidemic model in Section 5.3.

In our implementation, the history of the network evolution is stored as a list of `networkx.Graph` objects. This discrete-time representation allows for the calculation of temporal metrics using a sliding window approach, capturing the trajectory of node importance over time.

5.2 Centrality Computation

We analyze the importance of nodes not just by their instantaneous connectivity, but by the stability and momentum of their connectivity changes over time.

Comparison Framework We compute three primary metrics for each node i at time t to evaluate different aspects of temporal influence:

1. *Instantaneous Degree* ($k_i(t)$): the standard degree centrality in the snapshot G_t , representing immediate connectivity.
2. *Average Degree in Window* ($\bar{k}_i(t)$): a smoothed metric that aggregates the node’s importance over the recent history.

3. *Temporal Momentum Centrality* ($M_i(t)$): an ad-hoc metric introduced to explicitly capture the *trend* of a node’s connectivity evolution. Functioning similarly to a temporal derivative, this metric aims to distinguish whether a node is acquiring or losing contacts during the SIR-driven rewiring process, providing a directional perspective on centrality that complements static measures.

Analysis of Temporal Effects The temporal metrics are calculated using a rolling window of length τ (parameter `window_size`).

The *Average Degree* is computed as the arithmetic mean of the node’s degree over the window defined by the interval $[t - \tau + 1, t]$, effectively smoothing out high-frequency fluctuations:

$$\bar{k}_i(t) = \frac{1}{\tau} \sum_{j=0}^{\tau-1} k_i(t-j) \quad (2)$$

The *Temporal Momentum Centrality*, $M_i(t)$, is designed to project a node’s connectivity potential by combining its current state with its dynamical trend. We first define the smoothed rate of change $\Delta k_i(t)$ as the linear slope over the window:

$$\Delta k_i(t) = \frac{k_i(t) - k_i(t - \tau + 1)}{\tau} \quad (3)$$

The momentum metric is then constructed as a linear projection:

$$M_i(t) = \max(0, k_i(t) + \alpha \cdot \Delta k_i(t)) \quad (4)$$

where α is a scaling factor (measured in time steps) that weights the influence of the trend. Conceptually, this can be interpreted as a first-order Taylor expansion forecasting the node’s degree at time $t + \alpha$. The $\max(0, \cdot)$ operator ensures non-negativity, preserving physical interpretability. The function `calculate_rolling_temporal_metrics` implements this logic using vectorized operations over the collected graph snapshots.

Motivation for Temporal Momentum Centrality In an adaptive network driven by SIR dynamics, the instantaneous degree $k_i(t)$ can be ambiguous. Two nodes might share the same connectivity at a specific snapshot, yet represent fundamentally different roles in the network evolution: one might be an emerging hub attracting susceptible neighbours, while the other might be an infected node being actively isolated.

To illustrate the motivation of the *Temporal Momentum* metric (M_i), consider a toy example with two nodes, A and B, at time t . Both have an instantaneous degree of $k(t) = 10$, making them indistinguishable by static measures. However, their recent history differs significantly:

- Node A (Rising): had $k_A(t-1) = 5$. It is rapidly gaining connections.

- Node B (Falling): had $k_B(t-1) = 15$. It is rapidly losing connections.

As shown in Table 3, while $k(t)$ fails to separate them and $\bar{k}(t)$ might paradoxically favour the declining node due to lag, $M(t)$ correctly identifies the directional trend.

Node	Past $k(t-1)$	Current $k(t)$	Average $\bar{k}(t)$	Momentum $M(t)$
Node A (Rising)	5	10	7.5	+5 (Gaining)
Node B (Falling)	15	10	12.5	-5 (Losing)

Table 3. Toy Example of Metric Divergence. Both nodes appear identical in the current snapshot ($k = 10$). The Momentum Centrality is the only metric capable of immediately capturing the opposite dynamical trajectories.

This directional information is particularly relevant in our SIR-based rewiring model, where the loss of edges is a specific reaction to infection. The Momentum metric allows us to test the hypothesis that the derivative of centrality provides earlier warning signals of topological shifts compared to absolute values.

5.3 Epidemic Modelling on Temporal Networks

We simulate spreading processes using a modified Susceptible-Infected-Recovered (SIR) model that incorporates adaptive network topology changes.

SIR Model Nodes exist in one of three states: Susceptible (S), Infected (I), or Recovered (R). The dynamics are governed by two parameters:

- β : The infection probability. An interaction between an S node and an I node results in $S \rightarrow I$ transmission with probability β .
- γ : The recovery probability. An infected node transitions $I \rightarrow R$ with probability γ at each time step.

Epidemic Process Simulations Unlike static SIR models, our simulation incorporates an adaptive rewiring. This mechanism models the behavioural response of susceptible individuals attempting to avoid infection. The process, implemented in `simulate_SIR_with_rewiring`, proceeds as follows at each step t :

1. Infection and Recovery: Standard SIR transitions occur on the current edge set E_t .
2. Adaptive Rewiring (w): For every edge $(u, v) \in E_t$ where u is Susceptible and v is Infected, the edge is removed with probability w . This simulates S nodes breaking links with I neighbors to protect themselves.
3. Random Rewiring (p_{rand}): To maintain network density and simulate background noise, edges are randomly rewired with probability p_{rand} .

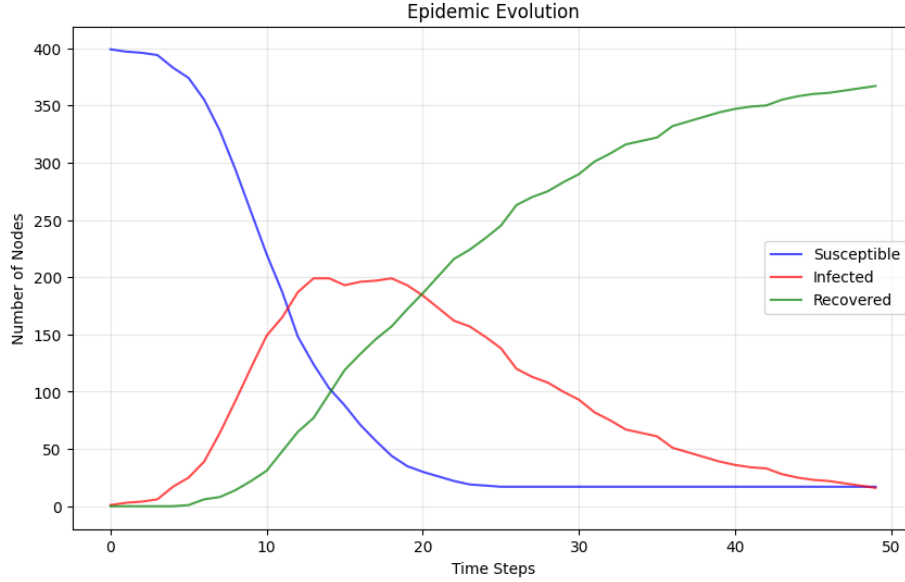


Fig. 4. Example of the dynamic SIR model behaviour

Comparison Framework To evaluate the impact of adaptive rewiring, we utilize a Monte Carlo simulation (implemented in `run_single_pair_simulation`). For each Monte Carlo iteration:

1. A random "Patient Zero" is selected.
2. Two parallel simulations are executed starting from the exact same initial graph structure and Patient Zero:
 - Scenario A: The network topology remains fixed ($w = 0$, $p_{rand} = 0$).
 - Scenario B: The adaptive rewiring mechanism is active ($w > 0$).

This testing approach ensures that observed differences in epidemic outcomes are solely attributable to the rewiring dynamics rather than initial condition variability.

Analysis of Spreading Dynamics and Mean-Field breakdown To investigate the interplay between topology and spreading, we perform a comparative analysis of the Effective Reproduction Number $R(t)$. Specifically, we contrast the realized spreading efficiency observed in simulations against the theoretical prediction derived from the Heterogeneous Mean-Field (HMF) framework.

The theoretical baseline is calculated under the "annealed network" assumption [1], which posits that the network is well-mixed at every time step and that the spreading potential depends solely on the instantaneous moments of the degree distribution. Under this approximation, the theoretical reproduction

number is given by:

$$R_{HMF}(t) = \frac{\beta}{\gamma} \left(\frac{\langle k^2(t) \rangle}{\langle k(t) \rangle} - 1 \right) \quad (5)$$

This formula assumes that the probability of infection is proportional to the global connectivity patterns (mass action law).

We compare this macroscopic prediction with the *Microscopic Effective* $R(t)$, computed by explicitly tracking the lineage of infections in the agent-based simulation. By averaging the number of secondary infections generated by all nodes infected at time t , we can quantify the deviation from the mean-field prediction, thereby isolating the suppressive effect of the adaptive "susceptible avoidance" mechanism that the HMF theory fails to capture.

5.4 Experimental Design for Testing Correlation between epidemic size and Temporal Metrics (H6)

To investigate the relationship between the dynamic evolution of node centrality and the final epidemic outcome, we designed a retrospective correlation analysis focused on the critical regime of the adaptive network. The experiment was conducted at a fixed transmission-to-recovery ratio of $\beta/\gamma = 0.7$. This operating point was selected based on preliminary analysis (see Figure 3), which indicated that at this threshold, the system is in a highly sensitive state where small variations in the local topological dynamics can lead to divergent epidemic sizes.

Since the network topology in an adaptive model evolves in response to the epidemic process itself, temporal metrics cannot be defined a priori as static predictors. Instead, this study treats them as concomitant variables tracked during the simulation to assess how strongly the realized centrality trajectory of a seed node correlates with the magnitude of the outbreak it generates.

Node Selection and Stratified Sampling To ensure the results are generalizable across the network's heterogeneity, we employed a stratified sampling strategy to select 50 distinct seed nodes. The network nodes were binned by degree, and candidates were drawn proportionally from each bin. This stratification ensures that the selected seeds represent the full spectrum of connectivity, from peripheral low-degree nodes to highly connected hubs, preventing bias towards specific topological classes.

5.5 Simulation Protocol and Metric Tracking

For each of the 50 selected seed nodes, we executed 100 independent Monte Carlo simulations of the SIR-Adaptive process. Each simulation proceeded until the epidemic reached its peak prevalence. During each run, we recorded the final epidemic size (defined as the maximum number of simultaneously infected and recovered nodes) and computed the three distinct centrality metrics discussed above for the seed node: To analyze the sensitivity of these metrics to the observation timescale, we repeated the analysis for three distinct window

sizes: $W \in \{3, 6, 9\}$ time steps. These values were chosen relative to the typical epidemic peak time observed in the system ($t_{peak} \approx 10$). In instances where the peak occurred earlier than the window size ($t_{peak} < W$), the window was truncated to the maximum available duration $[0, t_{peak}]$.

5.6 Statistical Analysis for H6 testing

The analysis was performed in two stages: a node-level correlation analysis followed by a global meta-analysis.

Within-Node Correlation For every seed node i , we calculated the Spearman rank correlation coefficient (ρ_i) between each temporal metric and the epidemic size across the 100 stochastic realizations. This non-parametric approach was chosen to capture monotonic relationships without assuming linearity. Simulations resulting in invalid correlations (e.g., due to zero variance in low-degree nodes) were excluded, resulting in a final effective sample of $N = 32$ validated nodes.

Meta-Analysis and Significance Testing To determine if a systematic relationship exists across the entire network, we treated the set of correlation coefficients $\{\rho_1, \rho_2, \dots, \rho_{32}\}$ as a distribution of effect sizes. We applied four distinct statistical tests to assess whether the mean correlation significantly deviated from zero:

- **Shapiro-Wilk Test:** To check the normality of the distribution of correlation coefficients.
- **One-Sample T-Test:** To test significance under the assumption of normality ($H_0 : \mu_\rho = 0$).
- **Wilcoxon Signed-Rank Test:** A non-parametric alternative for non-normal distributions.
- **Permutation Test:** A resampling method (10,000 permutations) to empirically estimate the p-value without distributional assumptions.

Statistical significance was defined at $\alpha = 0.05$. This multi-test approach ensures that our conclusions regarding the predictive power of temporal metrics are statistically robust and not artifacts of specific distributional assumptions.

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A Implementation details

This section documents how the code is organized and executed. Model definitions (temporal network representation, centrality definitions, and SIR+rewiring dynamics) are described in the Methods section (see Section 5 and the epidemic simulation description in Section 5.3). Here we focus on the software structure and the experiment pipeline.

A.1 Code organization and module boundaries

The implementation is split into four Python scripts under `src/` plus a single orchestration notebook:

- `src/Epidemic.py`: epidemic state-transition primitives used by the simulator.
- `src/Simulations.py`: simulation drivers (single-run) and Monte Carlo experiment runners.
- `src/Metrics.py`: post-processing utilities to compute analysis metrics from simulation history.
- `src/Plots.py`: plotting helpers used to generate the figures shown in the report.
- `CSN_2526_Project.ipynb`: entrypoint that sets parameters, runs experiments, and calls plotting/metric routines.

The modules exchange a `networkx.Graph` (with node epidemic states) and a `history` object returned by the simulator. The internal meaning/definition of temporal snapshots and sliding windows is covered in Methods (Section 5); in the implementation, `history` is treated as the single source for metric computation and plotting.

A.2 `src/Epidemic.py`

`Epidemic.py` provides the operations for updating node states (S/I/R) according to the SIR dynamics and for exposing helpers that the simulation layer can call. The goal is to keep state-transition logic isolated from:

- network evolution logic (rewiring), and
- experimental aggregation logic (Monte Carlo loops).

The epidemiological rules themselves are defined in Section 5.3.

SIR Implementation and Complexity Analysis

Initialization and Setup

Algorithm 1 Dynamic Network SIR: Initialization

Require: G_{orig} (Graph), β (Infection Prob), γ (Recovery Prob), w (Adaptive Rewire), p_{rand} (Random Rewire), T_{max}
Ensure: History of graph states $\mathcal{H}$

- 1: $G \leftarrow \text{DeepCopy}(G_{orig})$
- 2: $\mathcal{I} \leftarrow \{u \in V \mid \text{state}(u) = 'I'\}$
- 3: $\mathcal{R} \leftarrow \{u \in V \mid \text{state}(u) = 'R'\}$
- 4: $\mathcal{H} \leftarrow \emptyset$
- For** $t \leftarrow 1$ **to** T_{max}
- 5: Append G to $\mathcal{H}$
- 6: **if** $\mathcal{I} = \emptyset$ **then**
- 7: **break**
- 8: **end if**

Complexity Analysis

- **State Archiving:** The `copy.deepcopy(G)` inside the loop is an $O(N + M)$ operation, performed at every time step to preserve history.
- **Set Construction:** Creating sets $\mathcal{I}$ and $\mathcal{R}$ initially is $O(N)$.

Phase 1: Global Random Rewiring

Algorithm 2 Global Random Rewiring

- 1: **for all** $u \in V$ **do**
- 2: **if** $\text{random}() < p_{rand}$ **and** $\text{deg}(u) > 0$ **then**
- 3: $v \leftarrow \text{SelectRandomNeighbor}(u)$
- 4: $G.E \leftarrow G.E \setminus \{(u, v)\}$
- 5: $k \leftarrow \text{SelectRandomNode}(V \setminus \{u, \text{neighbors}(u)\})$
- 6: $G.E \leftarrow G.E \cup \{(u, k)\}$
- 7: **end if**
- 8: **end for**

Complexity Analysis

- **Operations:** Iterates over all N nodes.
- **Cost:** Selecting a random neighbour is $O(1)$. Finding a non-neighbour target involves a **while** loop. For sparse graphs ($M \ll N^2$), the probability of collision is low, making the cost amortized $O(1)$.
- **Complexity:** $O(N)$.

Phase 2: Adaptive Rewiring

Algorithm 3 Adaptive Rewiring

```

1: for all  $u \in V$  such that  $state(u) = 'S'$  do
2:    $\mathcal{N}_{inf} \leftarrow \{v \in neighbors(u) \mid state(v) = 'I'\}$ 
3:   for all  $v \in \mathcal{N}_{inf}$  do
4:     if  $random() < w$  then
5:        $Cands \leftarrow \{k \in V \mid k \neq u, (u, k) \notin G.E, state(k) \in \{'S', 'R'\}\}$ 
6:       if  $Cands \neq \emptyset$  then
7:          $G.E \leftarrow G.E \setminus \{(u, v)\}$ 
8:          $k \leftarrow \text{SelectRandom}(Cands)$ 
9:          $G.E \leftarrow G.E \cup \{(u, k)\}$ 
10:      end if
11:    end if
12:  end for
13: end for

```

Complexity Analysis

- The Python list comprehension `[node for node in G.nodes() if ...]` iterates at most over all N nodes in the graph to find a valid target.
- **Cost:** This scan occurs every time a specific edge is chosen for rewiring.
- **Complexity:**

$$O(N_{susceptible} \cdot k_{inf} \cdot N)$$

Phase 3: SIR Dynamics and Finalization

Algorithm 4 SIR Dynamics

```

1:  $NewInf \leftarrow \emptyset, NewRec \leftarrow \emptyset$ 
2: for all  $u \in \mathcal{I}$  do
3:   for all  $v \in neighbors(u)$  do
4:     if  $state(v) = 'S'$  and  $random() < \beta$  then
5:        $NewInf \leftarrow NewInf \cup \{v\}$ 
6:     end if
7:   end for
8:   if  $random() < \gamma$  then
9:      $NewRec \leftarrow NewRec \cup \{u\}$ 
10:  end if
11: end for
12: Update states in  $G$  using  $NewInf$  and  $NewRec$ 
13: Update sets  $\mathcal{I}$  and  $\mathcal{R}$  End For{End of main simulation loop}
14: return  $\mathcal{H}$ 

```

Complexity Analysis

- **Operations:** Iterates over infected nodes and their neighbours.
- **Complexity:** $O(N_{inf} \cdot k_{avg})$. In the worst case (full infection), this approaches $O(M)$.

Total Complexity Summary The overall time complexity is dominated by the behaviour observed in Phase 2:

$$O(T \cdot (N_{\text{susceptible}} \cdot k_{\text{inf}} \cdot N + M))$$

A.3 src/Simulations.py

`Simulations.py` is the execution layer. It contains:

- a single-run simulator (`simulate_SIR`) that returns `history`, and
- Monte Carlo wrappers used in the hypothesis-testing parts of the notebook.

Single run simulation by the notebook

```
history = simulate_SIR(...)
```

Implementation-wise, `simulate_SIR` is responsible for iterating time steps, applying epidemic transitions (delegated to `Epidemic.py`) and applying rewiring, while recording the time series required by `Metrics.py` and `Plots.py`. The model semantics are described in Section 5.3; the key note is that `simulate_SIR` produces a reusable trace for all downstream analysis.

Monte Carlo runners used by the notebook

- `run_targeted_analysis(...)` aggregates outcomes across a parameter sweep and across multiple stochastic runs, producing the tables/arrays used to generate Figure 2 and the associated comparisons.
- `run_monte_carlo_H5(...)` repeats simulations to estimate mean/std curves used in Figure 3.

Plotting is intentionally not performed inside these drivers: they return structured outputs which the notebook passes to `Plots.py` functions.

A.4 src/Metrics.py

`Metrics.py` implements the analysis layer: given `history` it computes the time series used to compare instantaneous degree, window-average degree, and the proposed momentum-like metric. Definitions and equations are in the Methods section (Section 5); this module implements them in a reusable way so that the notebook can:

- recompute metrics for different window sizes without rerunning the simulator, and
- extract per-node trajectories for visualization (e.g., Figure 1).

A.5 `src/Plots.py`

`Plots.py` contains the figure-building utilities used by the notebook. Concretely:

- `plot_evolution(history)` visualizes the epidemic evolution trace from a single run.
- `plot_full_comparison_row(...)` produces the multi-window metric comparison used for Figure 1.
- `plot_Rt_comparison_H5(...)` plots the mean/std curves produced by function `run_monte_carlo_H5` (Figure 3).

A.6 Notebook: `CSN_2526_Project.ipynb`

The notebook is the main entrypoint and performs:

1. parameter definition,
2. graph generation and initialization,
3. single-run simulation for qualitative inspection,
4. metric recomputation across window sizes,
5. Monte Carlo studies for the hypothesis tests, followed by plotting.

All model definitions referenced by these steps are documented in Methods (Section 5, Section 5.3); the notebook is primarily responsible for selecting parameter values and calling the correct driver functions.

Parameters and experiment grids This subsection lists the concrete parameter settings as they appear in the notebook.

Base network generation (fixed across experiments)

- `N = 400, m = 4` for `nx.barabasi_albert_graph(N, m)`.

Toy run (visualization only; not averaged)

- `steps = 50`
- `beta = 0.13, gamma = 0.08`
- `w = 0.3, p_rand = 0`

Window-size comparison

- `selected_nodes`: 8 uniformly sampled node indices.
- `max_window_size = len(history)//2` used as plotting bound; specific window sizes are generated internally by the plotting routine.

Monte Carlo grid for final-size comparisons

```
- max_steps = 2000
- num_sims = 100
- num_ratios = 30 (resolution of the swept transmission ratio grid used by
  run_targeted_analysis)
- analysis_ratio = 0.7
- windows = [3, 6, 9]
- gamma_val = 0.1 (passed as gamma to run_targeted_analysis)
- w = 0.3, p_rand = 0
```

Monte Carlo settings for $R(t)$ comparison

```
- steps = 40
- n_sims = 100
- beta = 0.13, gamma = 0.08
- w = 0.3, p_rand = 0.2
```